



UGANDA TECHNOLOGY AND MANAGEMENT UNIVERSITY
UTAMU

Examinations for January – April 2024 Semester

Course Code	CS 214		
Course Name	Practcum: Data Science		
Programme(s)	BSc.CS(Yr2,S2)		
School(s)	SCHOOL OF COMPUTING AND ENGINEERING		
Year of Study		Semester	
Examination Date	Tuesday-16th-April		
Starting Time	9:00 AM	Ending Time	12:00 PM

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Instructions:

1. This is an Open Book Examination
2. Attempt ALL Questions
3. Make sure you do all questions in one jupyter notebook. Also generate a PDF version of your notebook

GOOD LUCK!

QUESTION ONE: Basics (15 points)

- a. What skills do you think a good data scientist or someone good at data mining should have?
- b. What are some of the tasks you would ordinarily do in data wrangling?
- c. Explain one scenario in your past or future where you see data mining playing a key role.
- d. How else would you accomplish the same task without leveraging skills in data science?
- e. Why do you think data visualization is important?. Highlight a scenario where you would see data visualization playing a key role.
- f. Explain any 4 top trends in data science in 2024.

QUESTION TWO: Pandas Data Frames and Visualization (40 points)

Top 10 Premier League Table as of 15 April 2024

Club	MP	W	D	L	GF	GA	GD	Pts	Last 5
1  Man City	32	22	7	3	76	32	44	73	✓✓✓- -
2  Arsenal	32	22	5	5	75	26	49	71	✗✓✓- ✓
3  Liverpool	32	21	8	3	72	31	41	71	✗- ✓✓- -
4  Aston Villa	33	19	6	8	68	49	19	63	✓- ✗✓- -
5  Tottenham	32	18	6	8	65	49	16	60	✗✓- ✓✗
6  Newcastle	32	15	5	12	69	52	17	50	✓✓- ✓✗
7  Man United	32	15	5	12	47	48	-1	50	- - ✗- ✓
8  West Ham	33	13	9	11	52	58	-6	48	✗✓- ✗- -
9  Chelsea	30	12	8	10	55	52	3	44	- ✓- ✓- -
10  Brighton	32	11	11	10	52	50	2	44	- ✗- ✗✓

The above table has the first 10 premier league teams. Using Data Science knowledge. Do the following in jupyter notebook.

- Create the above table in a pandas data frame named **Pleague** with *columns* (**Club**, **MP**, **W**, **D**, **L**, **GF** and **GA** only). Do not mind the badges ignore them. Also neglect positions 1, 2 3, etc., they should be generated automatically. (5 points)
- Column **GD** should be added to the Pleague DataFrame with the formulae $GD = GF - GA$. (5points)
- Column **Pts** should be added to the Pleague DataFrame with the formulae $Pts = (W*3) + (D*1)$. (5 points)
- Display the first team. (3 points)
- Display the last team. (3 points)
- If the premier league ends today. Display the first 4 teams. (3 points)
- If the premier league ends today. Display the teams that will be relegated. i.e. the last three teams. (3 points)
- Display the 5th team. (3 points)
- Use seaborn to do any two visualizations that can explain insights from this data (the premier league table) (10 points)

QUESTION THREE: Data Extraction, Analysis and Visualization (45 points)

Use the KCSE Exam Results 2006 to 2010 dataset and do some processing on the data.

For each of the questions below, provide the requisite analysis from which you derive the answer.

- a. How many Males? How many Females are in the dataset? **(5 points)**
- b. Which school had the highest number of 'A's? **(5 points)**
- c. Which county had the lowest number of 'F's? **(5 points)**
- d. Which district had the highest number of 'B's on Average? What is that average number? **(5 points)**
- e. In which year did the schools in 'TAITA' have the highest number of 'C's? **(5 points)**
- f. Which school had the highest number of Females that got 'A's? In which year was this? **(5 points)**
- g. Plot a histogram of the average number of 'D+'s across the districts for the year 2006. **(5 points)**
- h. Use seaborn to do any two visualizations that can explain insights from this **(10 points)**